Lab 6

1. Objective

To configure a router to enable communication between:

- **Network A** → 3 PCs connected via Switch0
- **Network B** → 4 PCs connected via Switch1
- Both switches connected to Router1 (Cisco 2901)

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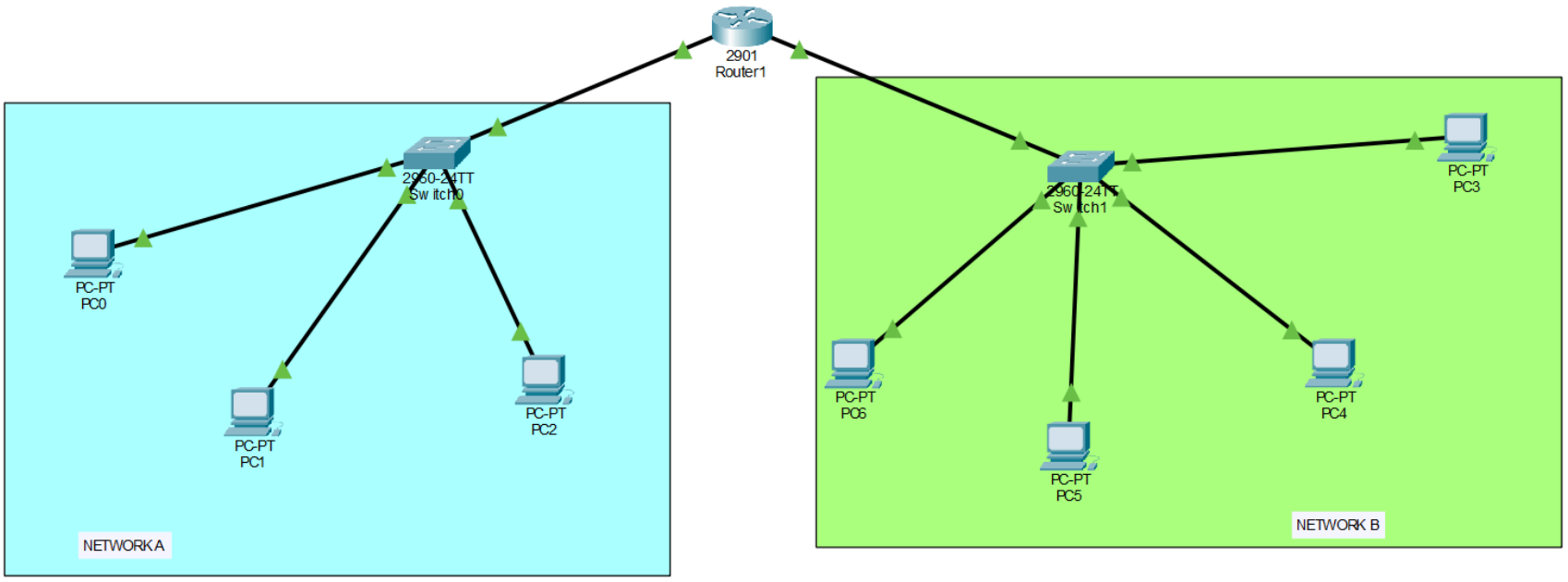
2. Network Design

2.1 Physical Topology



- Router: 2901
- Switch0 → Network A
- Switch1 → Network B
- Connected using copper straight-through wire

2.2 IP Addressing Scheme

Network A

- Subnet: **192.168.2.0/24**
- Default Gateway: **192.168.2.1**
- Example Host: PC1 → 192.168.2.3

Network B

- Subnet: **192.168.3.0/24**
- Default Gateway: **192.168.3.1**
- Example Host: PC4 → 192.168.3.4

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3. Router Configuration (CLI Output)

Router interfaces configured as:

- **GigabitEthernet0/0 → 192.168.2.1 255.255.255.0**
- **GigabitEthernet0/1 → 192.168.3.1 255.255.255.0**
- Both interfaces enabled using `no shutdown`

```
        --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEtherner 0/0
                                         ^
% Invalid input detected at '^' marker.

Router(config)#interface gigabitEtherner0/0
                                         ^
% Invalid input detected at '^' marker.

Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface gigabitEthernet0/1
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
```

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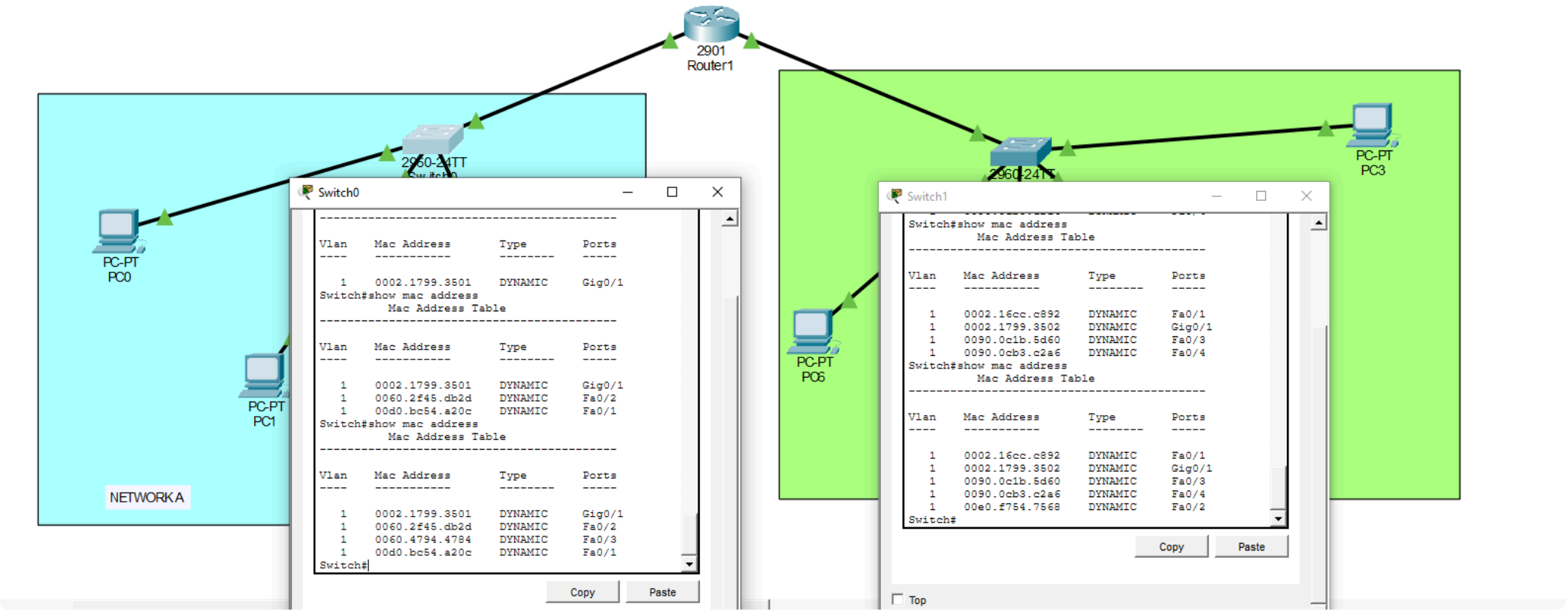
4. Switch MAC Address Tables

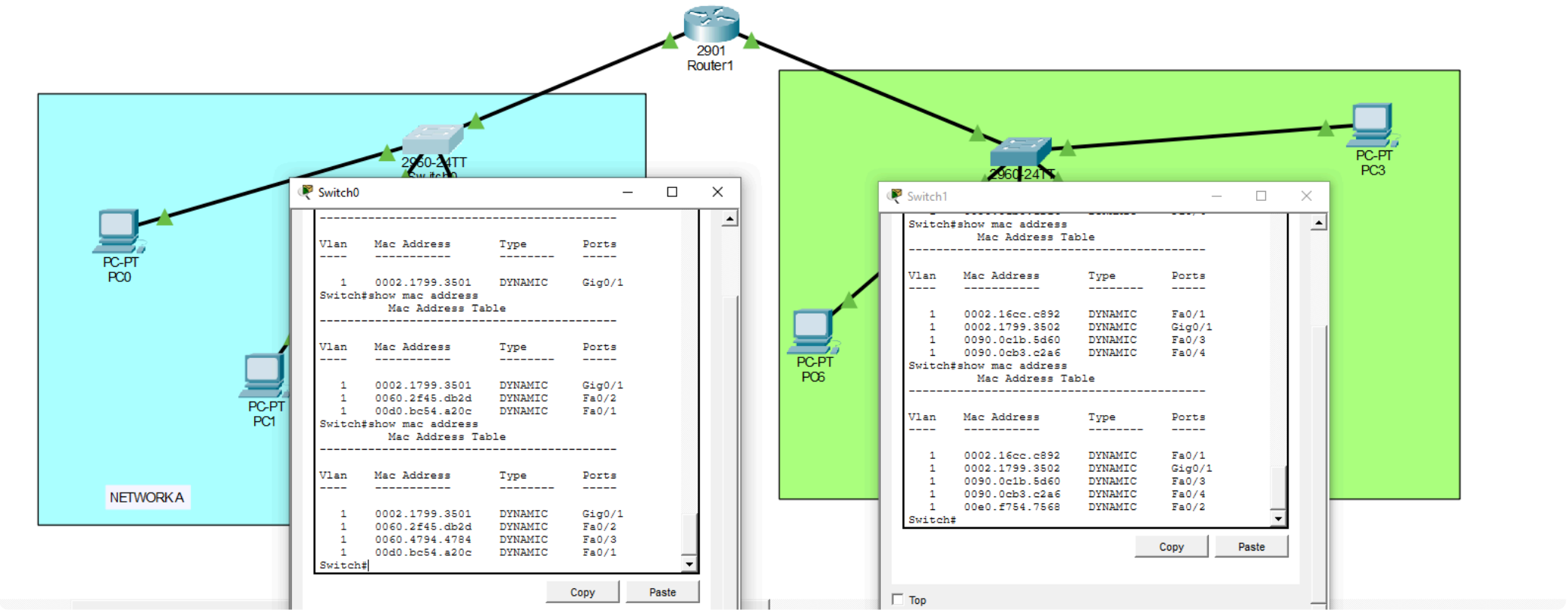
4.1 Switch0 (Network A)

- VLAN 1 dynamic MAC entries learned from connected PCs
- Ports: Fa0/1, Fa0/2, Fa0/3, Gig0/1



4.2 Switch1 (Network B)

- VLAN 1 dynamic MAC entries learned from connected PCs
- Ports: Fa0/1, Fa0/2, Fa0/3, Fa0/4, Gig0/1



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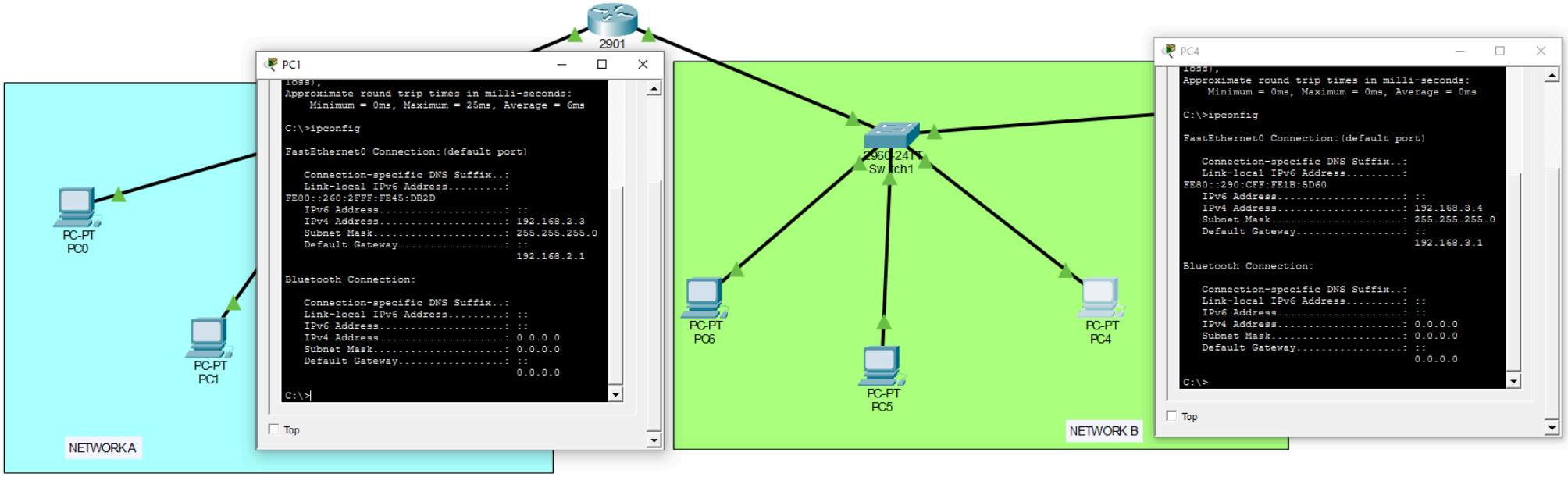
5. IP Configuration Verification

5.1 PC1 (Network A)

- IPv4 Address: 192.168.2.3
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

5.2 PC4 (Network B)

- IPv4 Address: 192.168.3.4
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.3.1



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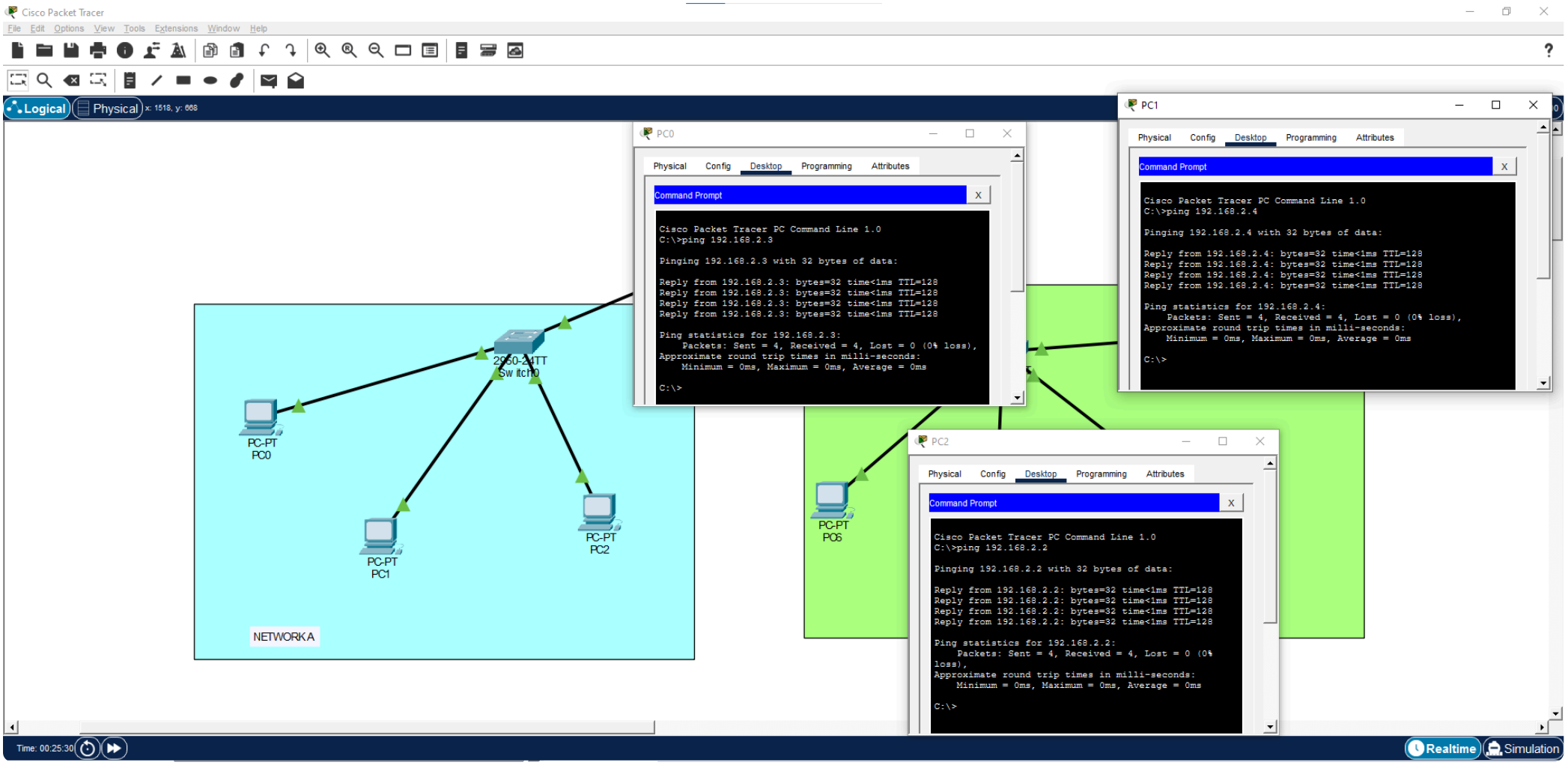
6. Connectivity Testing

6.1 Intra-Network Ping (Within Same Network)

Network A Example

PC2 → 192.168.2.3

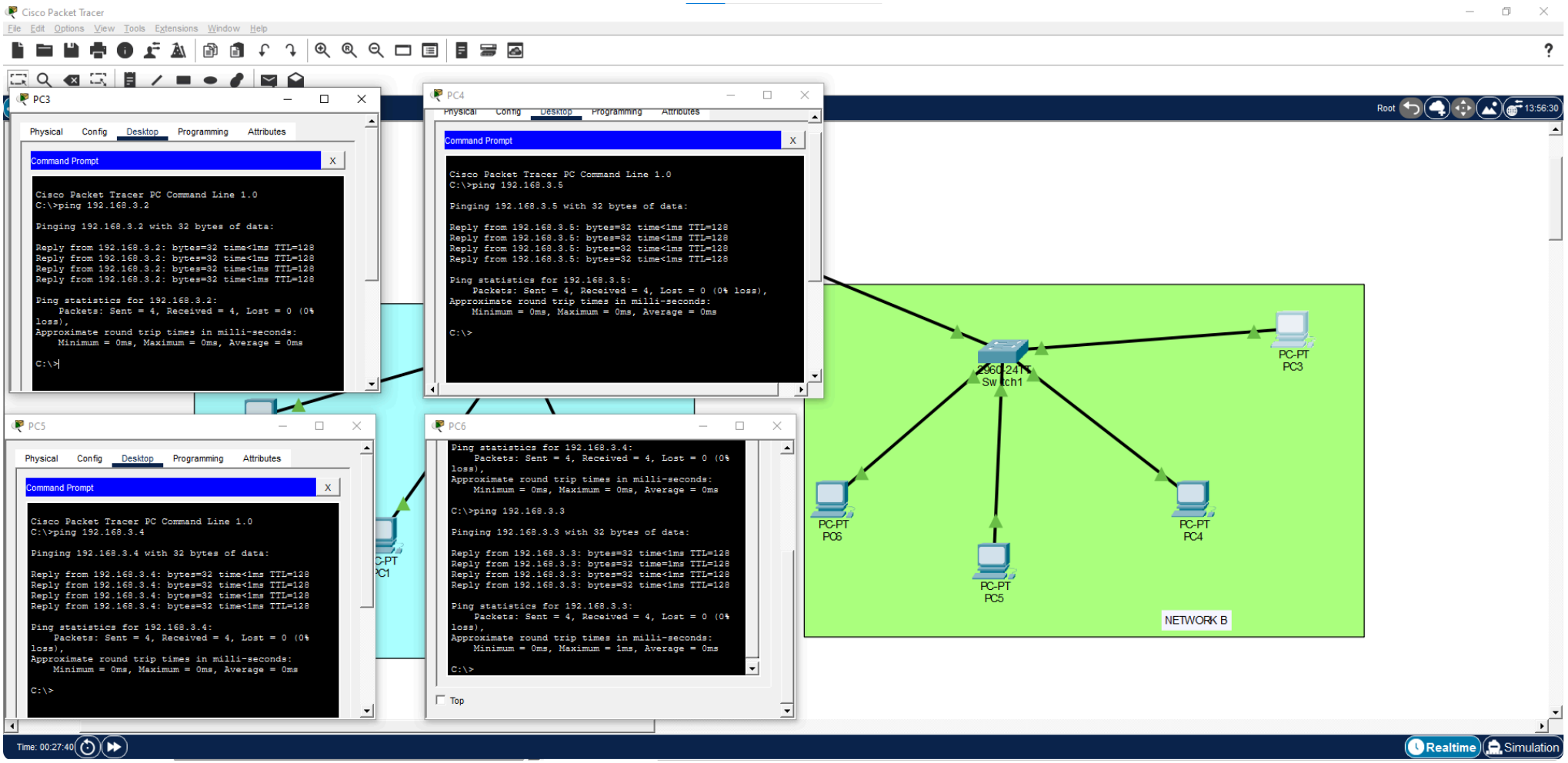
- Packets Sent: 4
- Packets Received: 4
- 0% Loss



Network B Example

PC3 → 192.168.3.5

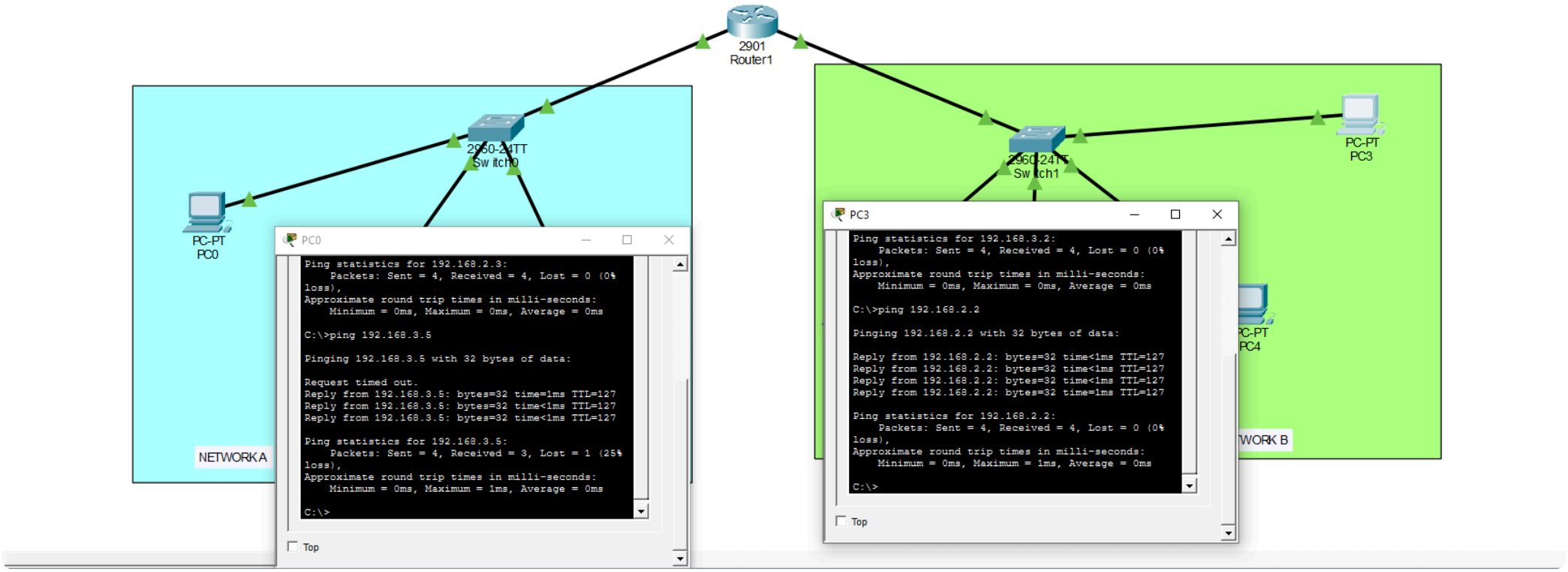
- Packets Sent: 4
- Packets Received: 4
- 0% Loss



6.2 Inter-Network Ping (Across Router)

Example:
PC0 (192.168.2.x) → PC3 (192.168.3.x)

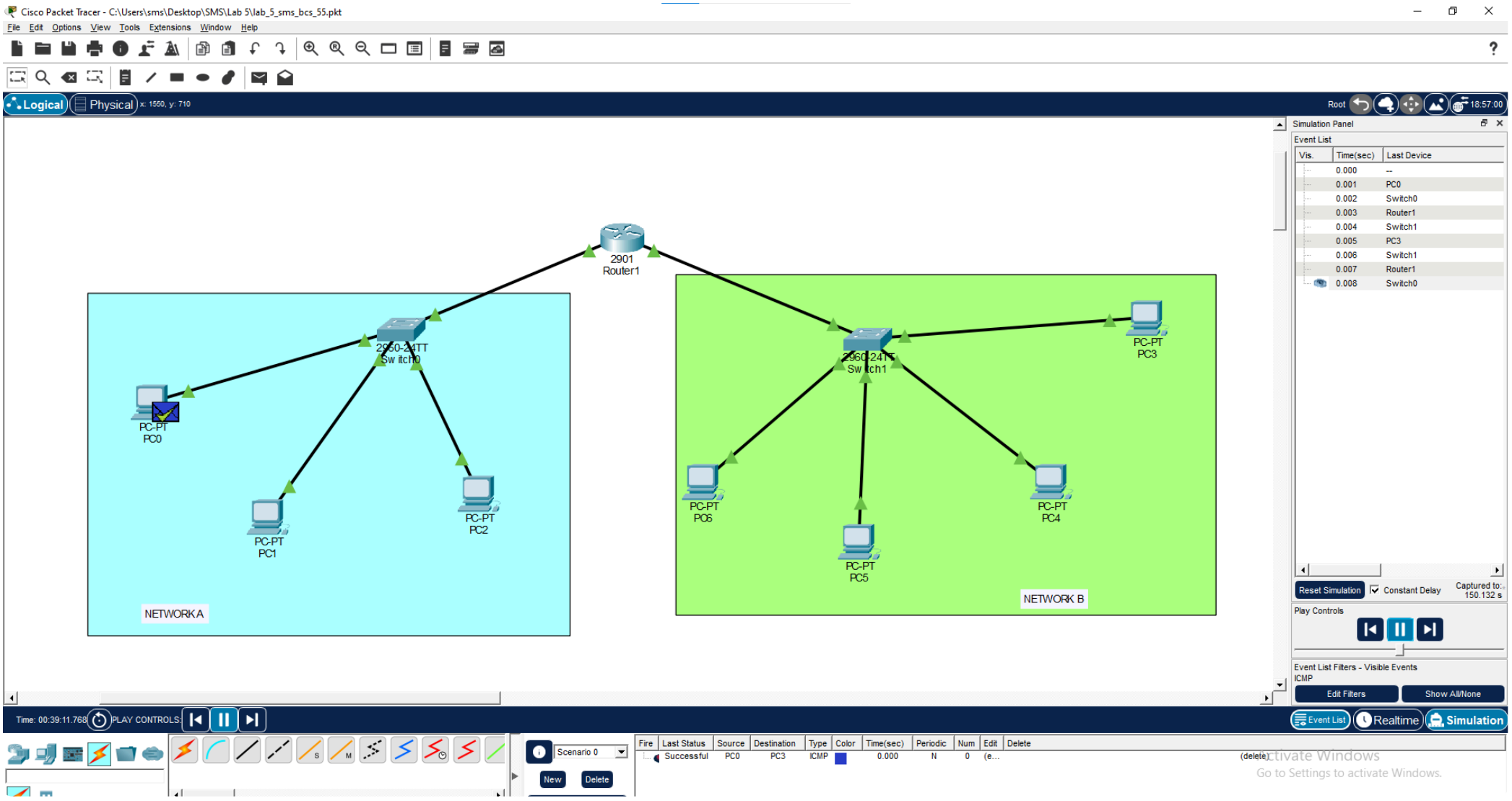
- Successful replies received
- Routing verified through Router1
- 0% packet loss after initial ARP resolution



7. Simulation Verification

Packet flow verified using Simulation Mode in Cisco Packet Tracer.

- ICMP packets successfully routed from Network A to Network B
- Event list confirms successful delivery



8. Conclusion

The router successfully enables communication between **Network A (192.168.2.0/24)** and **Network B (192.168.3.0/24)**.

- Switches correctly learned MAC addresses dynamically.
- IP addressing and default gateways configured properly.
- Intra-network and inter-network communication verified through successful ping results.

Inter-network routing is functioning as expected.

ID-Card

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